

Primitive Finite-Field Multipliers: Exact Orbits, Zeta and Koopman Determinants

Route-A structural certificate C172

Abstract

Let $Q \geq 2$ be any prime power and let a generate $\mathbb{F}_Q^\times$. Multiplication $T_a(x) = ax$ fixes zero and acts as one cycle of length $N = Q - 1$ on the nonzero field. We derive, uniformly in Q and a , every fixed-point count and the exact Artin–Mazur zeta $((1-z)(1-z^N))^{-1}$. The natural Koopman permutation is unitary; its spectrum is one extra eigenvalue 1 together with every N th root of unity, so its determinant is the reciprocal zeta. Inversion is an exact time reversor, and Koopman is self-adjoint exactly for $Q \leq 3$. These are all-parameter algebraic dynamics statements, not finite-field extrapolations.

Keywords: finite fields; primitive multipliers; permutation dynamics; Artin–Mazur zeta; Koopman operators; time reversal.

中文摘要

设 $Q \geq 2$ 为任意素数幂, a 为 $\mathbb{F}_Q^\times$ 的生成元。乘法映射 $T_a(x) = ax$ 固定零点, 并在非零域元素上形成长度 $N = Q - 1$ 的唯一周期。本文对任意 Q 与 a 推导全部不动点计数及精确 Artin–Mazur zeta 函数 $((1-z)(1-z^N))^{-1}$ 。自然 Koopman 置换算子是酉的; 其谱由一个额外特征值 1 与全部 N 次单位根组成, 故其行列式为该 zeta 函数的倒数。取逆映射给出精确时间反演, 而 Koopman 算子自伴当且仅当 $Q \leq 3$ 。这些结论是全参数代数动力学定理, 不是有限域样本外推。
关键词: 有限域; 本原乘子; 置换动力学; Artin–Mazur zeta; Koopman 算子; 时间反演。

1 Frozen field dynamics

Let $Q = p^e$, let a be primitive in $\mathbb{F}_Q^\times$, set $N = Q - 1$, and define $T_a(x) = ax$. One multiplication is one clock tick.

Theorem 1 (complete orbit law). Zero is fixed, $\mathbb{F}_Q^\times$ is one primitive N -cycle, and

$$\# \text{Fix}(T_a^n) = 1 + N \mathbf{1}_{N|n} = \begin{cases} Q, & N \mid n, \\ 1, & N \nmid n. \end{cases} \quad (1)$$

Consequently

$$\zeta_{T_a}(z) = \exp \left(\sum_{n \geq 1} \# \text{Fix}(T_a^n) \frac{z^n}{n} \right) = \frac{1}{(1-z)(1-z^N)}. \quad (2)$$

Proof. Every nonzero x has a unique form a^k with $k \in \mathbb{Z}/N\mathbb{Z}$, and $T_a(a^k) = a^{k+1}$. This proves the cycle and (1). Separating the fixed zero contribution gives

$$\sum_{n \geq 1} \frac{z^n}{n} + \sum_{m \geq 1} N \frac{z^{mN}}{mN} = -\log(1-z) - \log(1-z^N),$$

which proves (2) formally and for $|z| < 1$. □

2 Koopman certificate

On normalized counting $L^2(\mathbb{F}_Q)$, set $U_a f = f \circ T_a$.

Theorem 2. U_a is unitary and has one extra eigenvalue 1 from zero, together with every N th root of unity once from the nonzero cycle. Therefore

$$\det(I - zU_a) = (1-z)(1-z^N) = \zeta_{T_a}(z)^{-1}. \quad (3)$$

Proof. Composition with a permutation preserves normalized counting measure. The fixed point supplies one 1-eigenvector; discrete Fourier characters on the cyclic exponent coordinate diagonalize the long cycle. Their characteristic product is $1 - z^N$, yielding (3). □

3 Reversal and the self-adjoint boundary

Define $I(0) = 0$ and $I(x) = x^{-1}$ for $x \neq 0$. Then

$$IT_a I(x) = I(ax^{-1}) = a^{-1}x = T_a^{-1}(x), \quad (4)$$

with the same identity at zero. Hence $\Theta f(x) = \overline{f(Ix)}$ is a same-clock antiunitary reversal and $\Theta U_a \Theta^{-1} = U_a^{-1}$.

Theorem 3 (sharp boundary). U_a is self-adjoint if and only if $Q \leq 3$.

Proof. A unitary permutation is self-adjoint exactly when it equals its inverse, equivalently when every cycle has length at most two. The only nontrivial cycle here has length $N = Q - 1$, proving the claim. $\square$

At $Q = 2$, both field elements are fixed and (2)–(3) become $(1 - z)^{-2}$ and $(1 - z)^2$. At $Q = 3$, the nonzero cycle is a transposition. For every $Q \geq 4$, its length exceeds two and self-adjointness fails.

4 Exact validation boundary

The proof is independent of a polynomial representation of the field and covers every prime power. Exact ledgers for 18 fields, independent cycle enumeration, symbolic permutation matrices, byte replay and mutation attacks are regression sentinels rather than the proof. No target-spectrum or global arithmetic conclusion is asserted.